Technology Stack

Date	08 June 2026
Team ID	
Project Name	EcoTrack - Personal Carbon AI Footprint Tracker
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and

1. Frontend Presentation Layer:
   - Framework: React.js (Vite Build Tool)
   - Styling: Vanilla CSS & Tailwind CSS (Dark Mode / Glassmorphism)
   - Charts & Icons: Chart.js, React-Gauge-Chart, Lucide Icons

2. Backend Application Layer:
   - Web Server: Flask Python Web Server (Python 3.10+)
   - Modules: REST API Controllers, Cors Policy Handling, Dotenv Config

3. Database & Authentication Layer:
   - Database: Firebase Firestore Cloud Database

1	**Frontend / Client-Side**	React.js (Vite) & Tailwind CSS	Delivers rapid SPA rendering, dynamic glassmorphism UI layout, and interactive Chart.js carbon analytics.

2	**Backend / Server-Side**	Flask Web Server (Python 3.10+)	Provides lightweight REST API routing, CORS isolation, and instant Groq LLM API gateway integration.
3	**Database / Data Storage**	Firebase Firestore & Firebase Auth	Offers real-time cloud sync, flexible NoSQL document storage, and secure user session management.
4	**Cloud / Hosting / Deployment**	Vercel SPA Hosting & Gunicorn Server	Ensures 99.9% uptime, automated SSL certificates, and serverless global edge deployment.
5	**Version Control & CI/CD Pipeline**	Git, GitHub & GitHub Actions	Enables automated branch testing, code linting, and seamless team collaboration.

6	**Third-Party APIs & AI Integrations**	Groq LLaMA 3.3 API & Carbon Interface	Powers real-time AI lifestyle swap recommendations and provides accurate real-world CO2 emission factors.